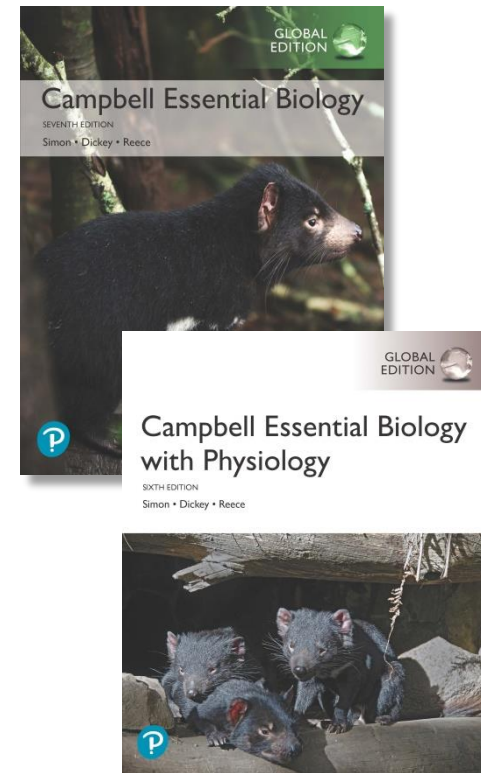


Campbell Essential Biology, Seventh Edition, Global Edition and Campbell Essential Biology with Physiology, Sixth Edition, Global Edition

Chapter 25 Hormones



PowerPoint® Lectures created by Edward J. Zalisko, Eric J. Simon, Jean L. Dickey, and Jane B. Reece

Biology and Society: Baseball's Ongoing Steroid Problem (1 of 2)

- Major League Baseball (MLB) has been plagued by widespread abuse of **anabolic steroids**, drugs known to enhance athletic performance.
 - During the so-called “steroids era” (from approximately the late 1980s to the late 2000s), MLB took no significant steps to prevent the use of anabolic steroids.
 - As a result, home run records were shattered, and fan interest in the game soared.
 - Major League Baseball banned anabolic steroids in 2001.

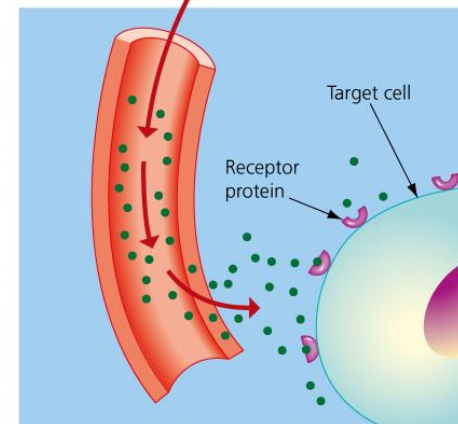
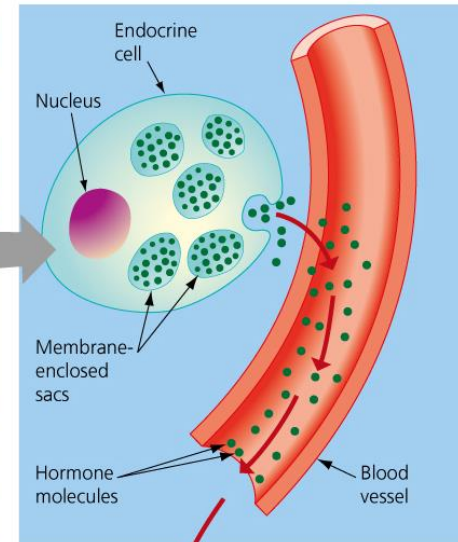
Biology and Society: Baseball's Ongoing Steroid Problem (2 of 2)

- Steroids are hormones, chemical signals that help regulate body functions.
 - **Anabolic steroids**, however, are synthetic hormones that mimic the male sex hormone **testosterone**.
 - There is disagreement about the severity of dangers posed by anabolic steroids.
 - When abused, they are associated with many health risks, such as aggression, psychiatric disorders, and kidney, liver, and cardiovascular damage.
 - But proponents point out that anabolic steroids are widely prescribed to very ill people, such as patients recovering from AIDS and cancer.

Hormones: An Overview (1 of 5)

- The diverse cells of your body are constantly **communicating** with one another, coordinated through chemical and electrical signals, traveling by way of two major organ systems.
 1. In the nervous system, the signals are **primarily electrical** and are transmitted via neurons.
 2. The **endocrine system** is a group of glands and tissues that produce and secrete chemicals to initiate and maintain body functions. **Hormones are chemical signals released by endocrine cells** and carried to all locations in the body, usually **via the blood**. **Figure 25.1** shows the release of hormone molecules from an endocrine cell.

Hormone secretion from an endocrine cell



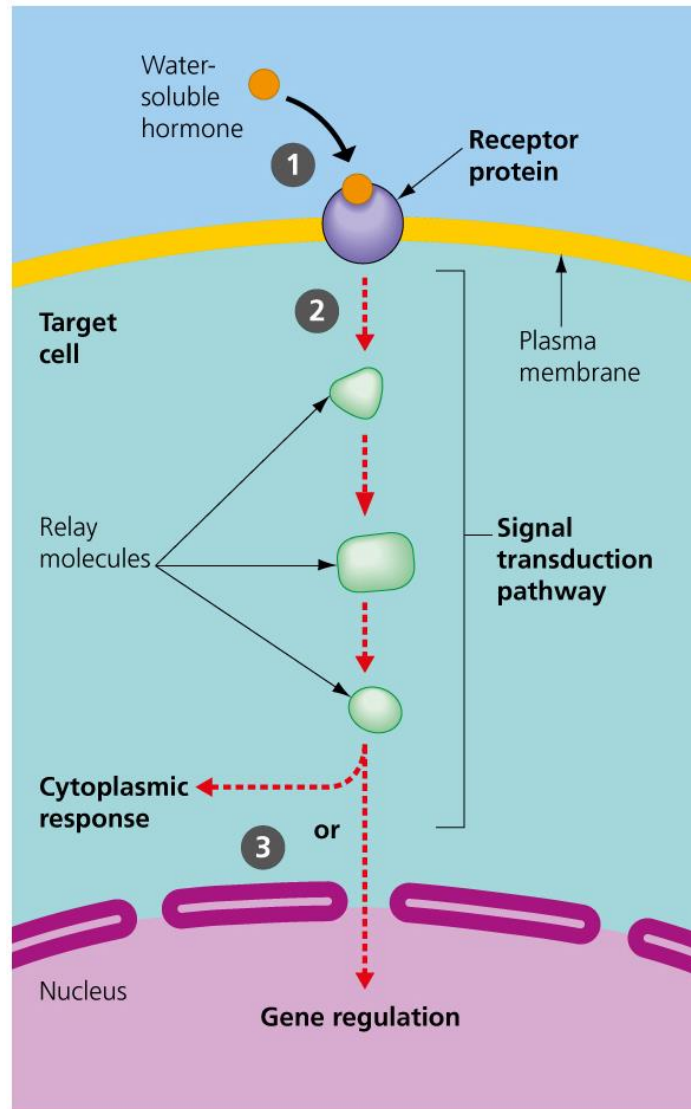
Hormones: An Overview (2 of 5)

- The circulatory system carries hormones throughout the body, but a hormone can only bind to **a target cell with receptors for that specific hormone**.
- Because hormones reach all parts of the body, the endocrine system is important in **controlling whole-body activities**.

Hormones: An Overview (3 of 5)

- Hormones trigger changes in target cells differently, depending on whether the hormone is water-soluble or lipid-soluble.
 - **Water-soluble hormones** trigger responses without entering the cell.
 - Water-soluble hormones cannot pass through the phospholipid bilayer of the plasma membrane, but they can bring about cellular changes without entering their target cells.
 - **Figure 25.2** shows how water-soluble hormones cause changes in target cells.

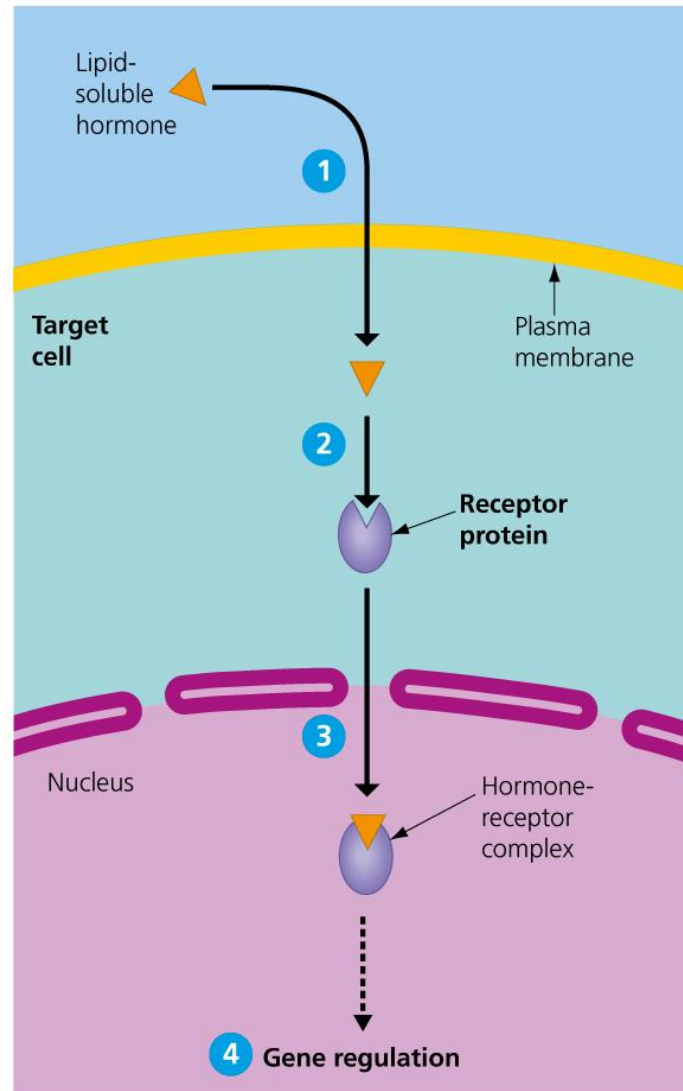
A hormone that binds to a plasma membrane receptor



Hormones: An Overview (4 of 5)

- In contrast, **lipid-soluble hormones** pass through the phospholipid bilayer and trigger responses by binding to receptors inside the target cell.
 - **Steroid hormones**—including natural ones such as the sex hormones (testosterone and estrogen) as well as artificial anabolic steroids—work in this manner.
 - **Figure 25.3** shows how lipid-soluble hormones cause changes in target cells.

A hormone that binds to an intracellular receptor



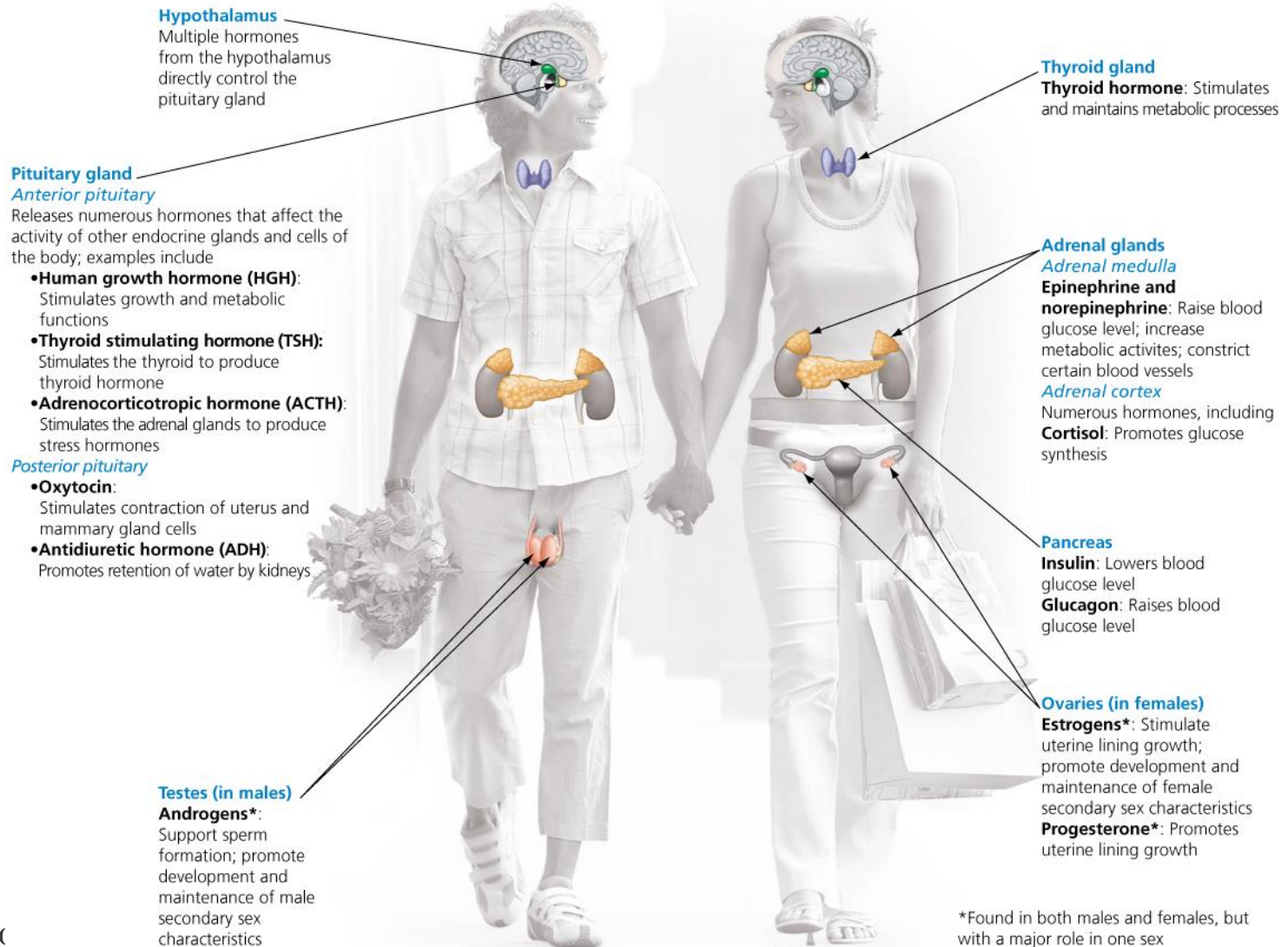
Hormones: An Overview (5 of 5)

- Animals have many hormones in common, a result of shared ancestry.
 - Yet over the course of evolution, the functions of a given hormone have diverged between species.
 - For example, thyroid hormone plays a role in regulating metabolism in many vertebrates, yet stimulates metamorphosis in frogs (the process in which tadpoles become adults).

The Human Endocrine System

- Hormones are made and secreted mainly by organs called **endocrine glands**.
 - The human endocrine system consists of about a dozen major glands. Some of these, such as the thyroid and pituitary glands, are endocrine specialists, secreting hormones into the blood.
 - Other organs, for example, the **stomach and the pancreas** of the digestive system, are primarily non-endocrine but have some cells that secrete hormones.
 - **Figure 25.4** shows the locations of some of the major human endocrine glands.

Endocrine glands in humans

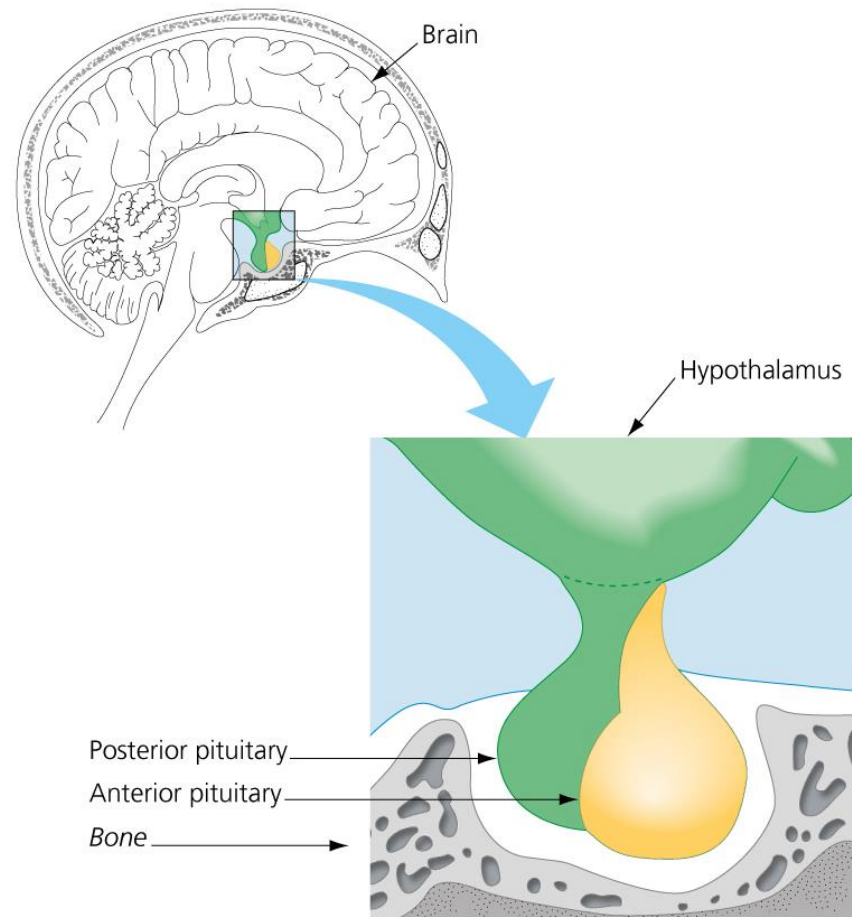


The Hypothalamus and Pituitary Gland

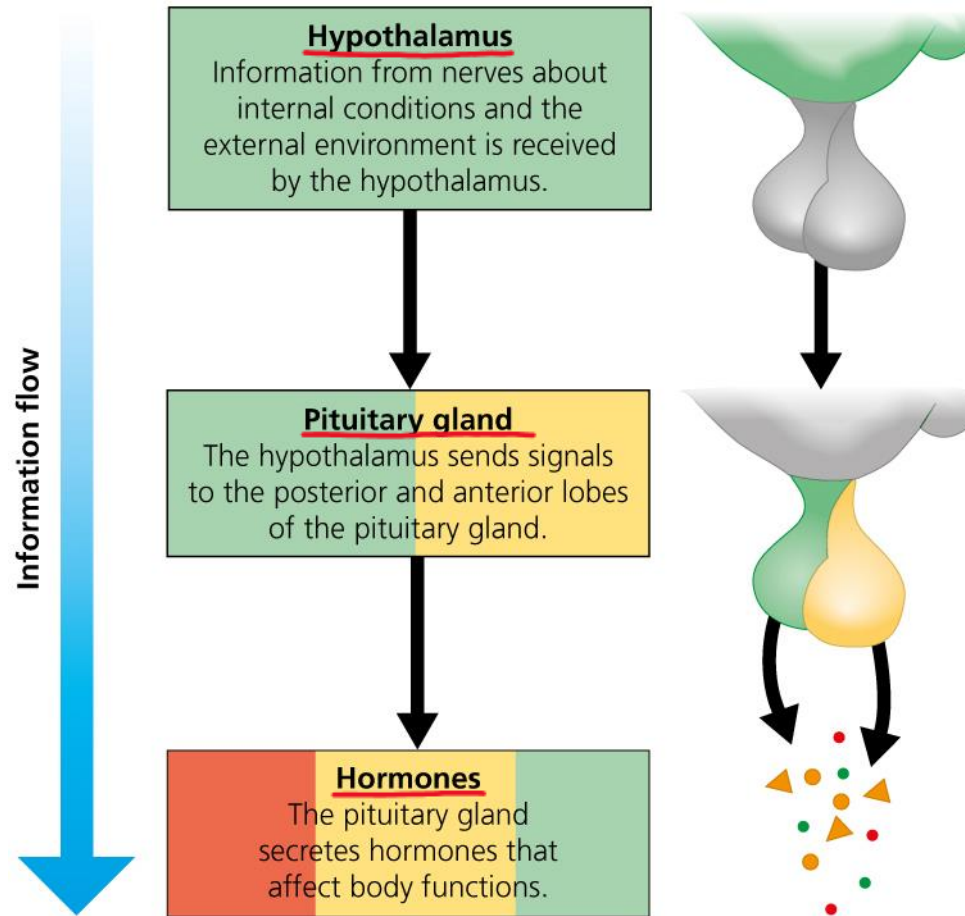
(1 of 4)

- The **hypothalamus**, a gland in the brain, acts as the main control center of the endocrine system.
 - It receives information from nerves about the internal condition of the body and about the external environment and then responds by sending out appropriate nervous or endocrine signals.
 - These signals directly control the **pituitary gland**, a pea-sized structure that hangs down from the hypothalamus.

Location of the hypothalamus and pituitary



Master control exerted by the hypothalamus

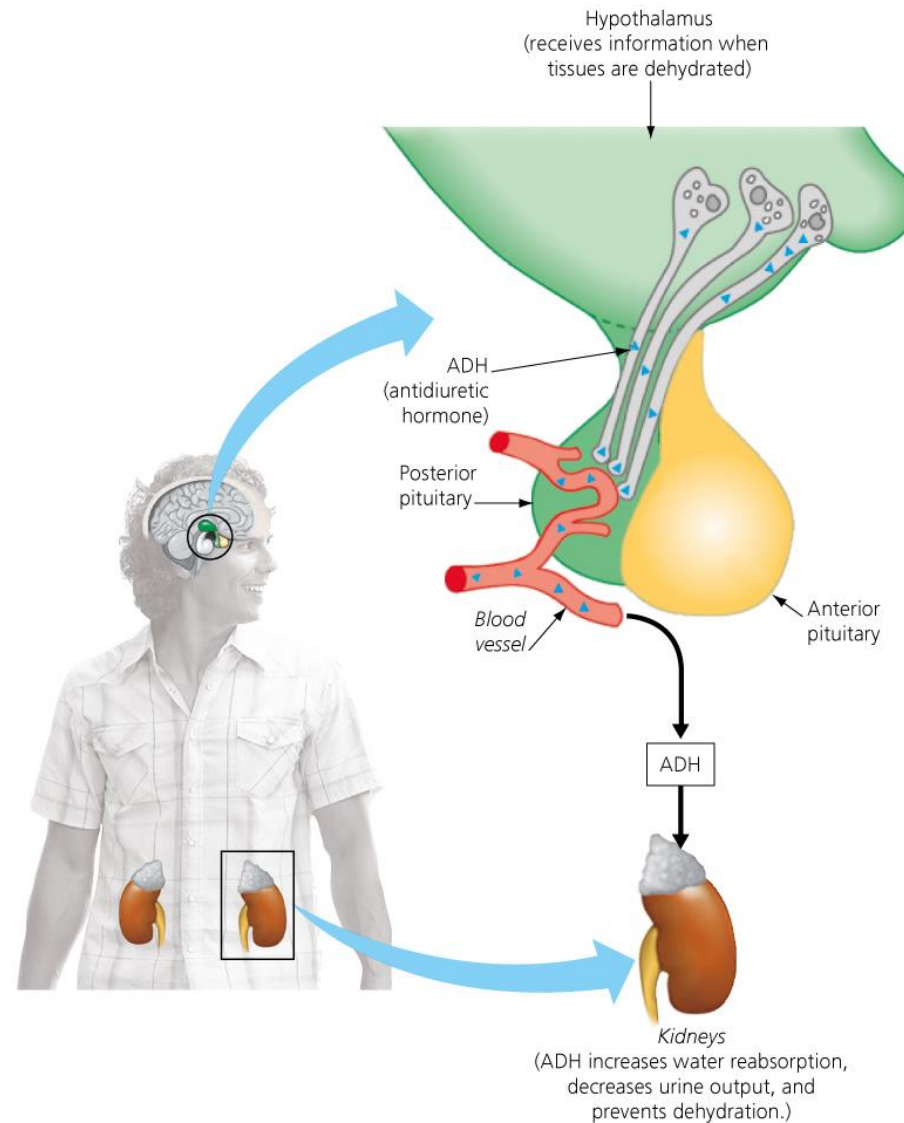


The Hypothalamus and Pituitary Gland

(2 of 4)

- The endocrine system maintains **homeostasis**. For example, when the hypothalamus receives information that the body tissues are dehydrated, it makes antidiuretic hormone (**ADH**, **Figure 25.7**).
 - ADH is stored and **released by the pituitary gland** and signals kidney cells to reabsorb more water so that less urine is produced and serious dehydration is prevented.
 - As an individual rehydrates, the hypothalamus slows the release of ADH from the pituitary and the kidneys increase the output of urine.

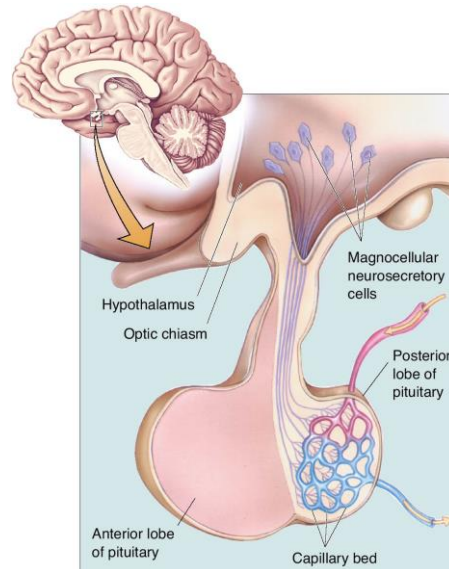
ADH and osmoregulation



The Hypothalamus and Pituitary Gland

(3 of 4)

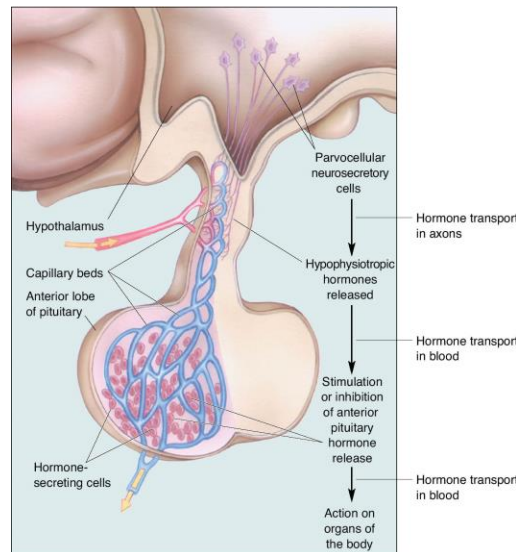
- The **pituitary gland** consists of two distinct parts.
 1. The **posterior pituitary** is an extension of the hypothalamus, **stores hormones**, and **secretes hormones made in the hypothalamus**. Two hormones are released by the posterior pituitary.
 - a) **ADH**, as just noted, participates in osmoregulation.
 - b) **Oxytocin** stimulates the contraction of the uterus during childbirth and causes the mammary glands to release milk.
 2. The **anterior pituitary** synthesizes and secretes **its own hormones** directly into the blood.



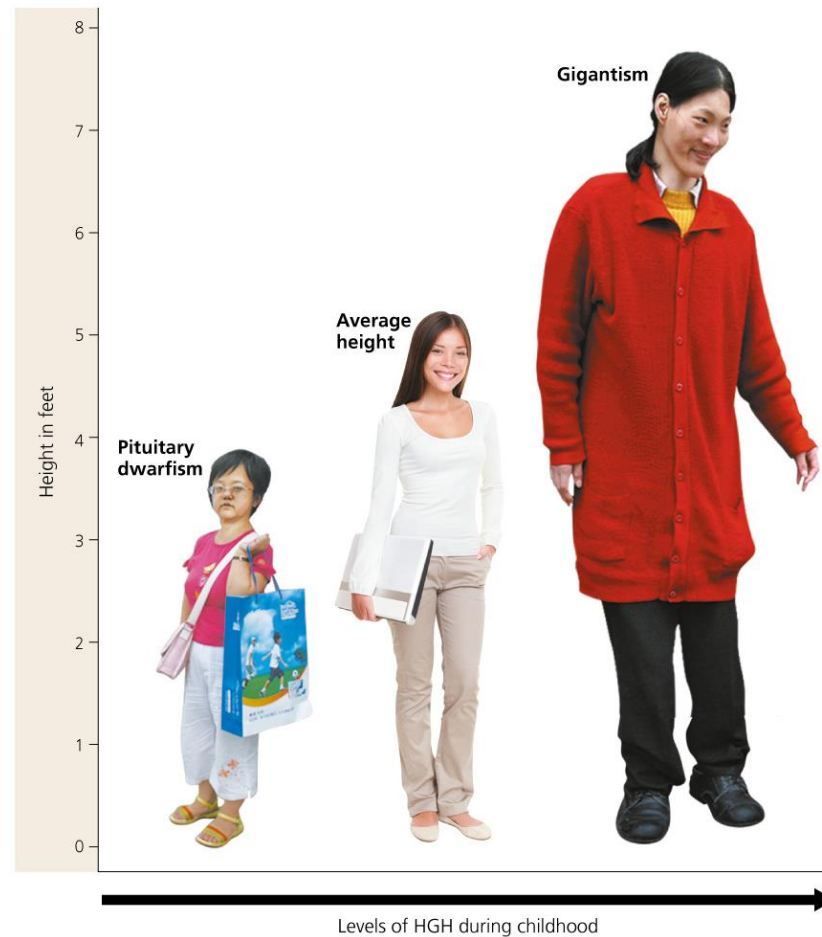
The Hypothalamus and Pituitary Gland

(4 of 4)

- The **hypothalamus** exerts control **over the anterior pituitary** by secreting two kinds of hormones into short blood vessels that connect the two glands.
 1. Releasing hormones stimulate the anterior pituitary to secrete specific hormones.
 2. Inhibiting hormones induce the anterior pituitary to stop secreting specific hormones.
- One of several anterior pituitary secretions is **human growth hormone (HGH)**, which promotes development and enlargement of all parts of the body. Too much HGH can produce gigantism. Too little HGH can lead to dwarfism.



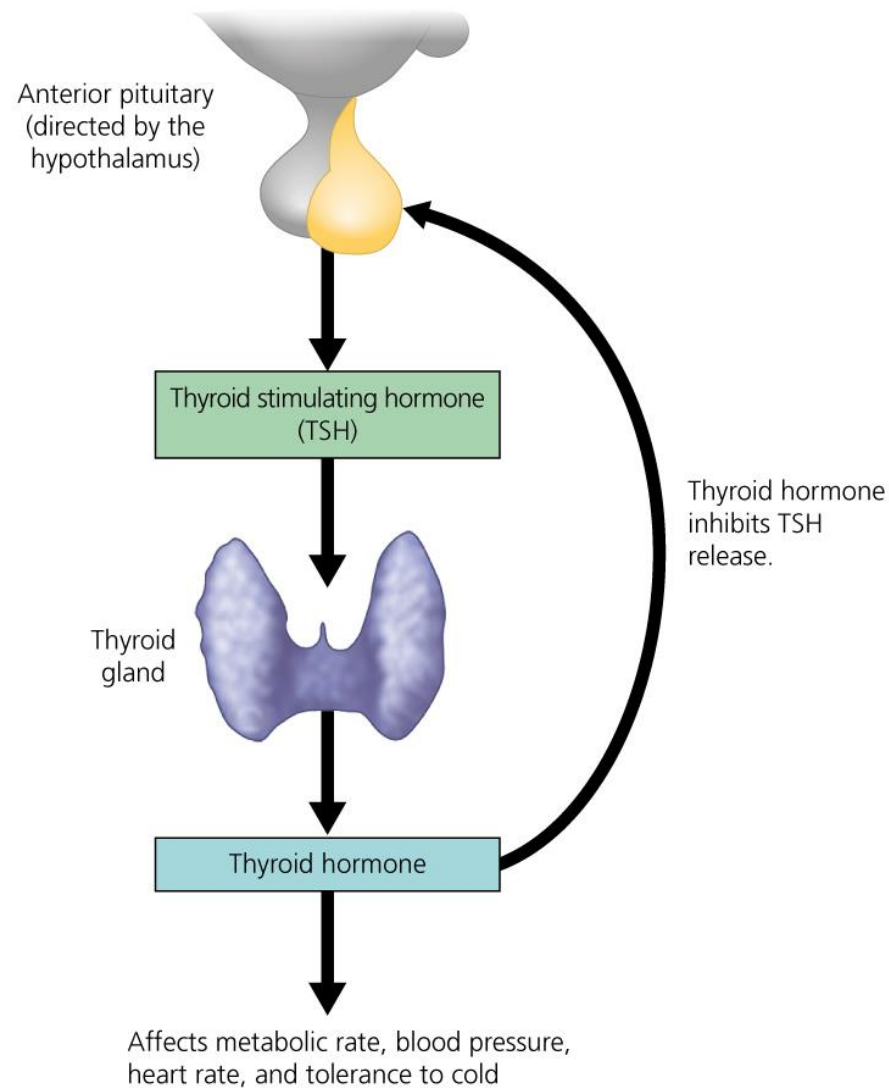
Pituitary growth hormone disorders



The Thyroid and Metabolism (1 of 5)

- The **thyroid gland** is located in the front of your neck and produces **thyroid hormone** in response to **thyroid stimulating hormone (TSH)** released from the anterior pituitary.
 - **Thyroid hormone** affects many functions of the body including metabolic rate (the amount of energy expended over a period of time), heart rate, blood pressure, tolerance to cold, and the development and maturation of bone and nerve cells in children.
 - **Figure 25.9** shows how thyroid production is regulated.

Thyroid hormone production



The Thyroid and Metabolism (2 of 5)

- An additive to table salt, **iodine (I)** is necessary for normal thyroid function. **Thyroid hormone cannot be synthesized without iodine.**
 - Without iodine, and thus a lack of thyroid hormone, **goiter** eventually develops.
 - Goiter still affects many people in less industrialized nations where iodine is not available in the diet.

The Thyroid and Metabolism (3 of 5)

- Disease or environmental toxins are causes of hypothyroidism.
 - **Hashimoto's disease** is an autoimmune disorder in which the immune system attacks thyroid cells.
 - Environmental toxins that adversely affect the endocrine system, termed **endocrine disruptors**, are commonly found in pesticides and plastic containers.
 - For example, BPA (bisphenol A) is an endocrine disruptor found in some plastic water bottles.
 - A growing body of evidence links particular endocrine disruptors to thyroid disease in animals.

The Thyroid and Metabolism (4 of 5)

- **Hyperthyroidism** (overproduction of thyroid hormone) is characterized by weight loss, high blood pressure, rapid heart rate, and increased sensitivity to heat.
 - It is treated by using low doses of radioactivity to kill some thyroid cells.
 - But high doses of radiation can be deadly.
 - Nuclear fallout often produces dangerous plumes of radioactive iodine that enter the air, water, and food supply.

Hyperthyroidism: 갑상선 기능 항진증 (갑상선 호르몬의 과다 분비)

Nuclear fallout: 핵폭발로 인한 방사능 낙진

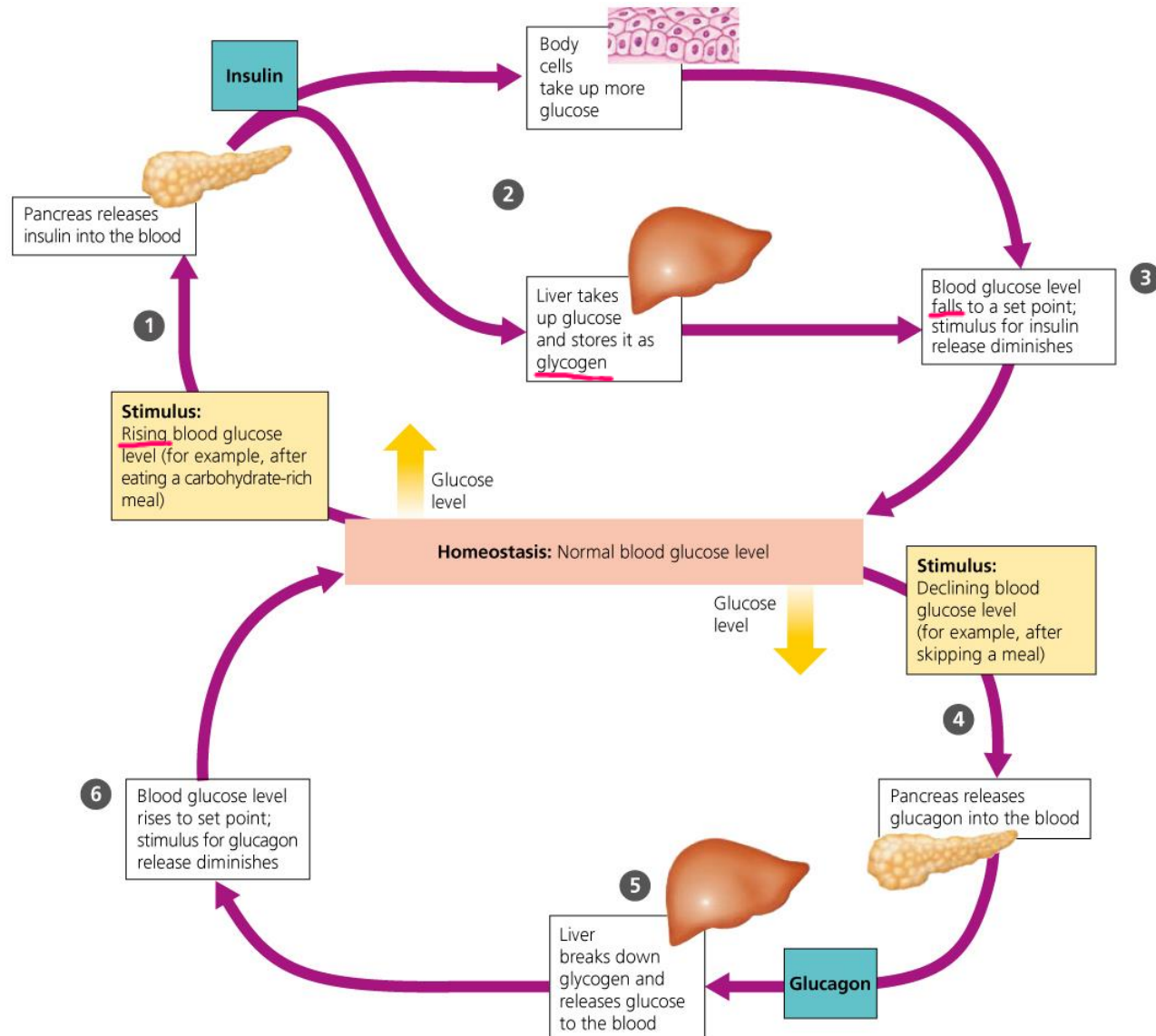
The Thyroid and Metabolism (5 of 5)

- When radioactive iodine accumulates in the thyroid, the risk of thyroid cancer increases. To help protect the thyroid, a stable form of iodine, potassium iodide (KI), is taken in pill form before or immediately after exposure to nuclear fallout.
- This treatment “fills up” the thyroid with stable iodine so that the thyroid cannot take up the harmful radioactive iodine.
- These pills are sometimes advertised to consumers as a necessity for “doomsday” preparation.

The Pancreas and Blood Glucose (1 of 4)

- The **pancreas** secretes two hormones, **insulin and glucagon**, that play important roles in managing the body's energy supplies.
 - Insulin and glucagon help maintain a balance between the amount of **glucose available in the blood** and the amount of glucose stored as the polymer glycogen in body cells.
 - As shown in **Figure 25.10**, insulin and glucagon are **antagonistic hormones**, countering each other in a feedback circuit that precisely manages the level of glucose in the blood.

Glucose homeostasis



The Pancreas and Blood Glucose (2 of 4)

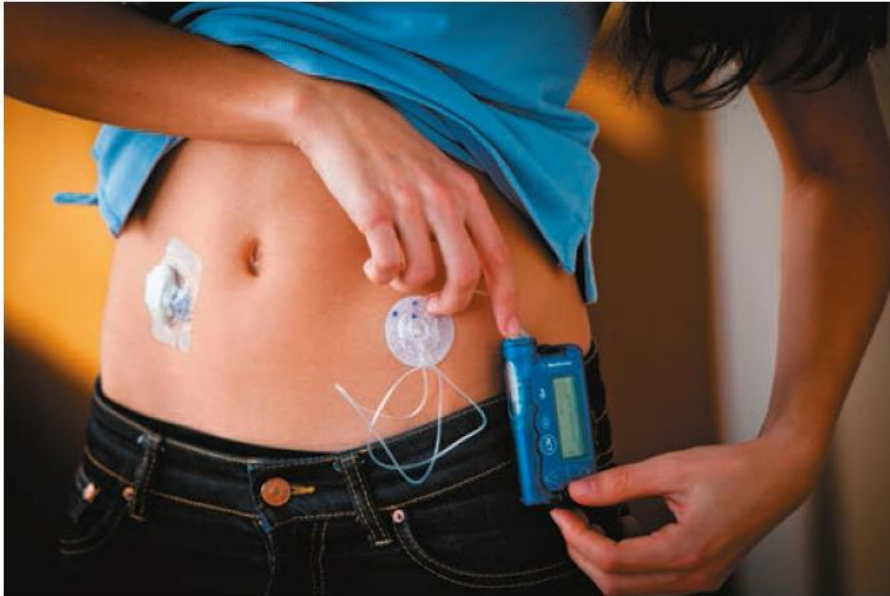
- **Diabetes mellitus** is a serious hormonal disease that affects about 1 in every 12 Americans. Millions of them do not even know they are ill. If current trends continue, 1 in every 3 Americans will be affected by 2050.
 - In diabetes, **body cells are unable to absorb glucose from the blood**, either because
 1. there is **not enough insulin in the blood** (as in **type 1, or insulin-dependent, diabetes**) or
 2. because the target cells **do not respond normally to the insulin in the blood** (as in **type 2, or non-insulin-dependent, diabetes**). Of those Americans who have diabetes, more than 90% have type 2.

The Pancreas and Blood Glucose (3 of 4)

3. A third type of diabetes, called **gestational diabetes**, can affect any woman during pregnancy, even one who has never shown symptoms of diabetes before.
 - There are treatments for diabetes but **no cure**.
 - Diabetics measure their blood glucose routinely throughout the day.
 - Type 1 patients require regular injections of insulin.
 - Type 2 is almost always associated with being **overweight and underactive**, although whether obesity causes diabetes remains unknown.

Gestational diabetes: 임신당뇨증

Measuring blood glucose



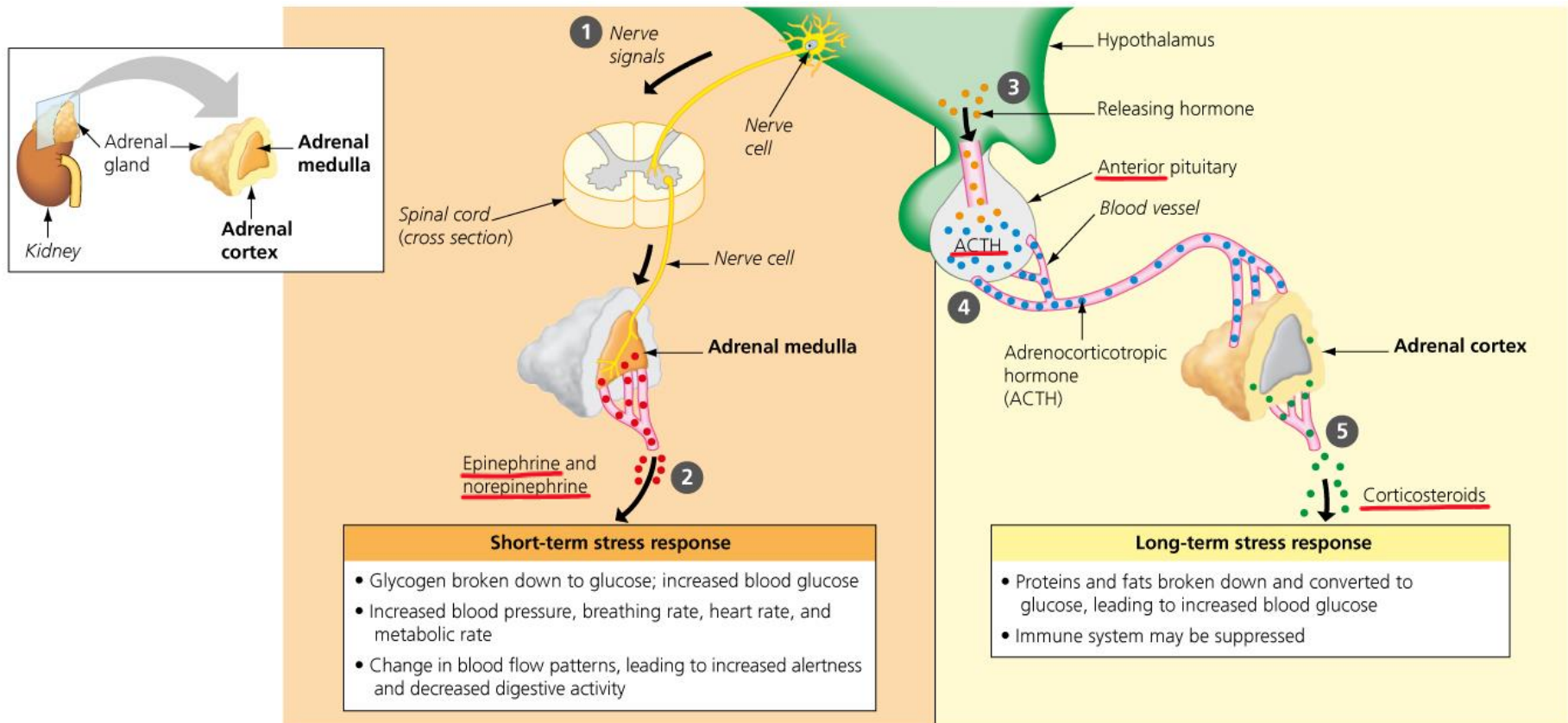
The Adrenal Glands and Stress (1 of 4)

- **Epinephrine** (also called adrenaline) and **norepinephrine** are “fight-or-flight” hormones, are water-soluble hormones, and can be activated in seconds and last for minutes or hours.
- The **adrenal glands** are paired structures resting atop each kidney. Each adrenal gland consists of two glands:
 1. the central **adrenal medulla** (which makes the “fight-or-flight” hormones) and
 2. the outer **adrenal cortex**.

The Adrenal Glands and Stress (2 of 4)

- Hormones secreted by the **adrenal cortex** can provide **a slower, longer-lasting response to stress** that can last for hours or days (as shown in the right half of **Figure 25.13**).
 - The hypothalamus secretes a releasing hormone that stimulates the pituitary to secrete a hormone called **ACTH (adrenocorticotrophic hormone)**.
 - ACTH stimulates cells of the **adrenal cortex** to synthesize and secrete a family of **lipid-soluble steroid hormones** called **corticosteroids**, which includes the hormone cortisol.

How the adrenal glands control our response to stress



The Adrenal Glands and Stress (3 of 4)

- **Cortisol** helps promote the synthesis of glucose from non-carbohydrates, such as proteins and fats.
 - The energy is needed when the body is coping with stressful situations.
 - As cortisol increases energy for vital functions, it can suppress less immediate functions, such as the body's immune system.

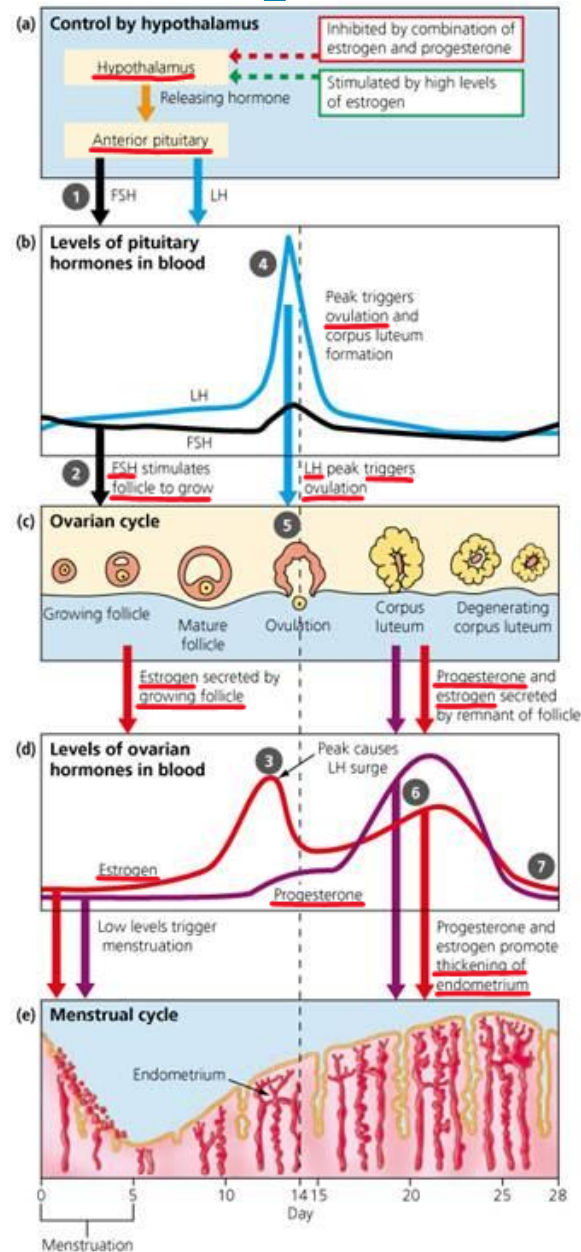
The Gonads and Sex Hormones (1 of 4)

- Women and men produce the same sex hormones.
- The **gonads**, or sex glands (ovaries in the female and testes in the male), produce three categories of sex hormones:
 - 1) estrogens, 2) progestin, and 3) androgens.
 - The proportions of these hormones are what differentiate the sexes.
 - The sex hormones are steroid hormones (**lipid-soluble**) that are secreted under the direction of the hypothalamus and anterior pituitary gland.

The Gonads and Sex Hormones (2 of 4)

- Growth and development are affected by sex hormones, as are the reproductive cycle and sexual behavior.
- **Estrogens and progestins** are found in higher concentrations in women than in men.
 - **Estrogens** maintain the female reproductive system and promote the development of specific female features, such as breasts and wider hips.
 - **Progestins**, such as **progesterone**, are primarily involved in preparing the uterus to support a developing embryo.
 - Both hormones are critical to the menstrual cycle.

The reproductive cycle of the human female



The Gonads and Sex Hormones (3 of 4)

- Men have high levels of **androgens** (the main one being **testosterone**), which stimulate the development and maintenance of the male reproductive system.
- **Endocrine disruptors** affect the thyroid, but others affect reproductive function by altering the balance of sex hormones.
 - In one study, scientists examined the effects of an endocrine disruptor on **frogs**. They raised male frogs in the absence (control) or presence of a chemical found in **weed-killers called atrazine**.

The Gonads and Sex Hormones (4 of 4)

- When male frogs reached adulthood, they found that atrazine-exposed males were far less successful at mating behaviors and had a lower testosterone level.
- Furthermore, 10% underwent complete sex reversal. They became females capable of producing eggs!

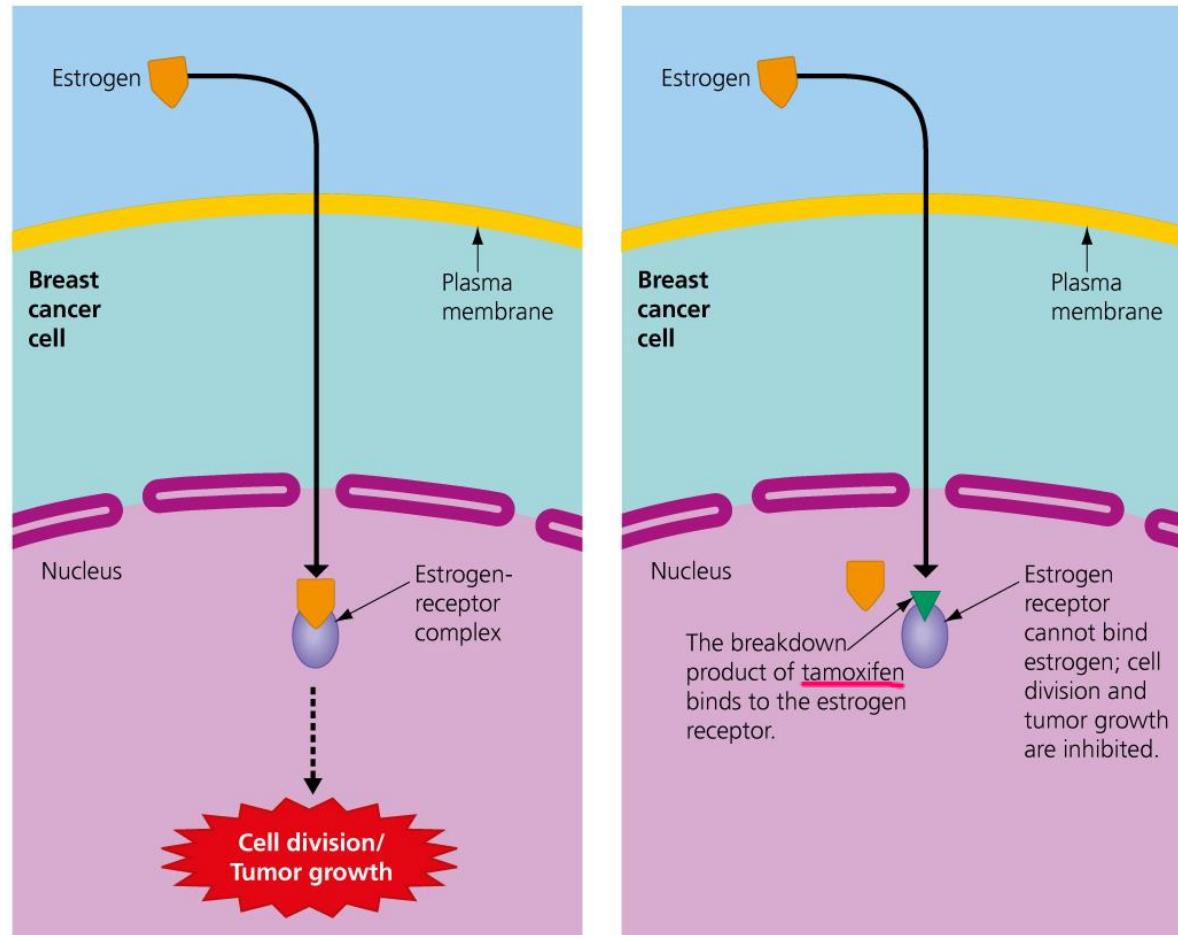
Mimicking Sex Hormones (1 of 2)

- Chemicals that affect the balance of sex hormones are not always detrimental.
 - A drug used in **breast cancer treatment** decreases estrogen's effects, and synthetic hormones are useful for boosting function in individuals deficient in particular hormones.
 - These chemicals have similarities in structure and function to the natural hormones they mimic.

Mimicking Sex Hormones (2 of 2)

- Approximately 70% of all breast cancers rely on estrogen to grow and divide. Inhibiting estrogen binding or its effects is a successful way to treat many breast cancers. Tamoxifen is a drug that inhibits estrogen. Once in the body, it brakes down into molecules that bind to estrogen receptors.
- As individuals age, sex hormone levels decline.
 - One treatment for the side effects of menopause is synthetic estrogen.
 - Anabolic steroids, artificial hormones that are similar in structure and function to testosterone, are sometimes used to treat the gradual decline of testosterone in men.

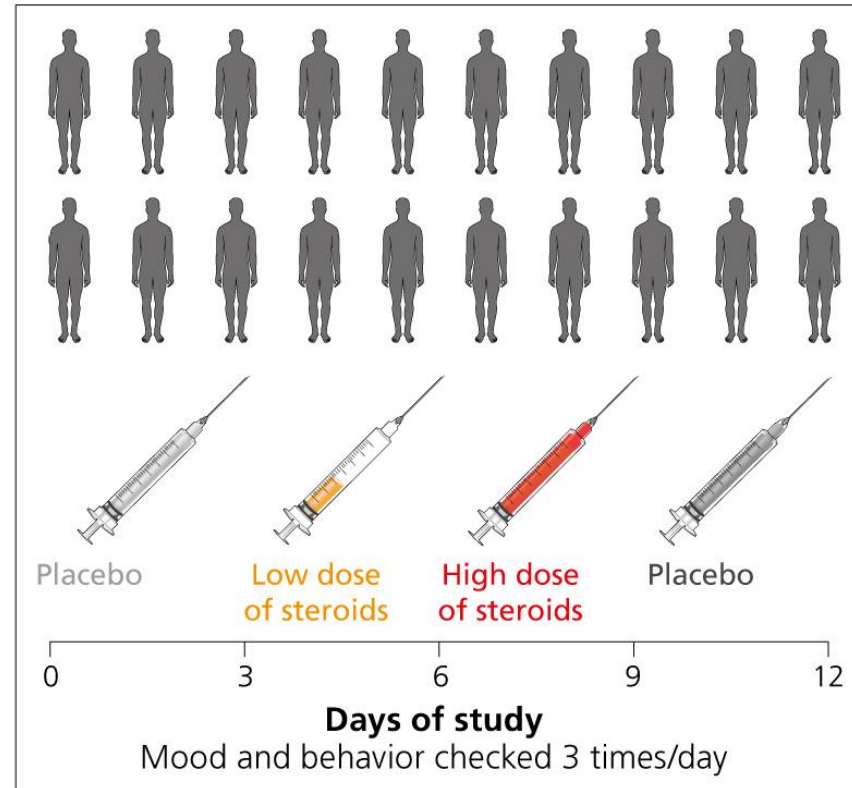
Breast cancer drug treatment



The Process of Science: Do ‘Roids Cause Rage?

- **Background:** The media has speculated about the link between abuse of anabolic steroids and violent mood swings, often called “roid rage”. How can scientists test to see if it is a real phenomenon?
- **Method:** A group of doctors at the National Institutes of Health investigated the relationship between anabolic steroids and mood in the lab.
 - The researchers hypothesized that increasing doses of steroids would produce measurable changes in behavior in a group of volunteers.

During the 12 day study, 20 volunteers were injected with varying doses of anabolic steroids



(a) During the 12 day study, 20 volunteers were injected with varying doses of anabolic steroids.

The Process of Science: Do ‘Roids Cause Rage (1 of 2)

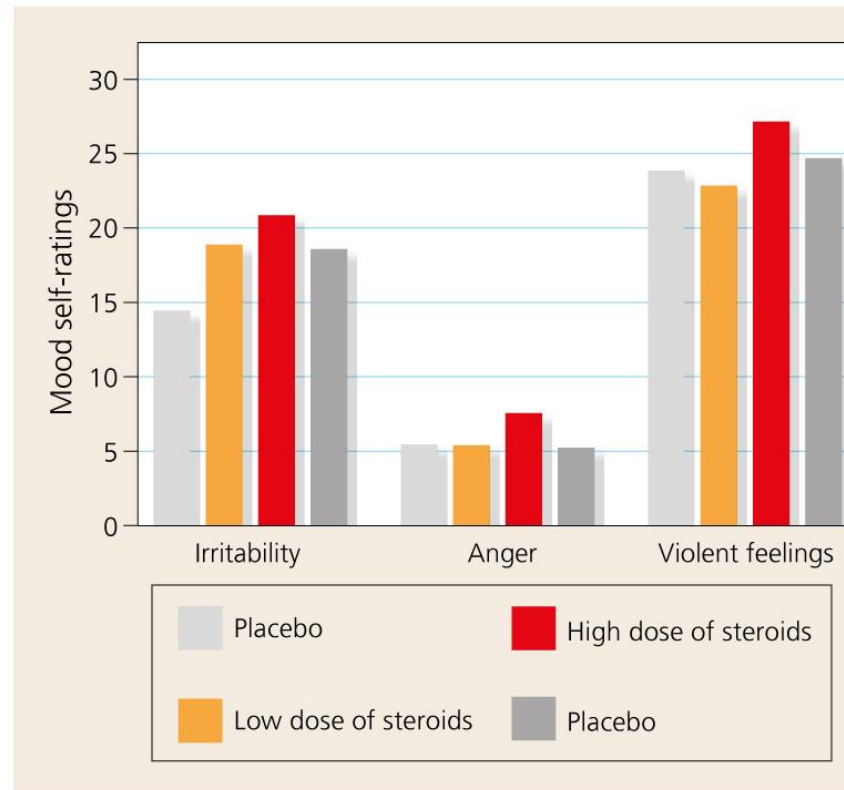
- The doctors recruited 20 male volunteers between the ages of 18 and 42, none of whom had a history of drug abuse or psychiatric problems.
- As shown in **Figure 25.15b**, the research participants were given a placebo drug (no steroids = days 1–3), a low dose of steroids (days 4–6), a high dose of steroids (days 7–9), and then a placebo (days 10–12). Throughout the study, the participants were observed continuously and questioned three times a day using standard psychiatric tests that measure mood and behavior.

The Process of Science: Do 'Roids Cause Rage? (2 of 2)

- **Results:** Their data revealed a significant increase in ratings for hostility, anger, and violent feelings during the high-dose period.
 - Even though the sample size was small, the researchers noted that their participants received doses well below the levels taken by some abusers.
 - Also, the participants used just a single drug (whereas some athletes abuse several at once), suggesting abuse could produce stronger mood swings. Together with many similar studies, the data suggest a link between the abuse of anabolic steroids and violent behavior.



Each set of bars in the graph represents one measure of mood



(b) Each set of bars in the graph represents one measure of mood. Each color represents one dosage state. Notice the increased scores during the high-dose period (the red bar in each set).

Evolution Connection: Steroids and Male Aggression (1 of 2)

- **Testosterone and other androgens** promote the development and maintenance of the human male reproductive system and play a similar role in all vertebrates.
 - In many vertebrates, they are responsible for **male vocalizations**, such as the territorial songs of birds and the mating calls of frogs.
 - Androgens are also widely associated with **aggressive behavior**. For example, in male elephant seals, androgens promote the development of bodies weighing more than 2 tons and aggressive behavior toward other males. These males fight by slamming their bodies against each other.

Male elephant seals battling for dominance



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